GH raisoni INTERNATIONAL SKILL TECH UNIVERSITY Pune

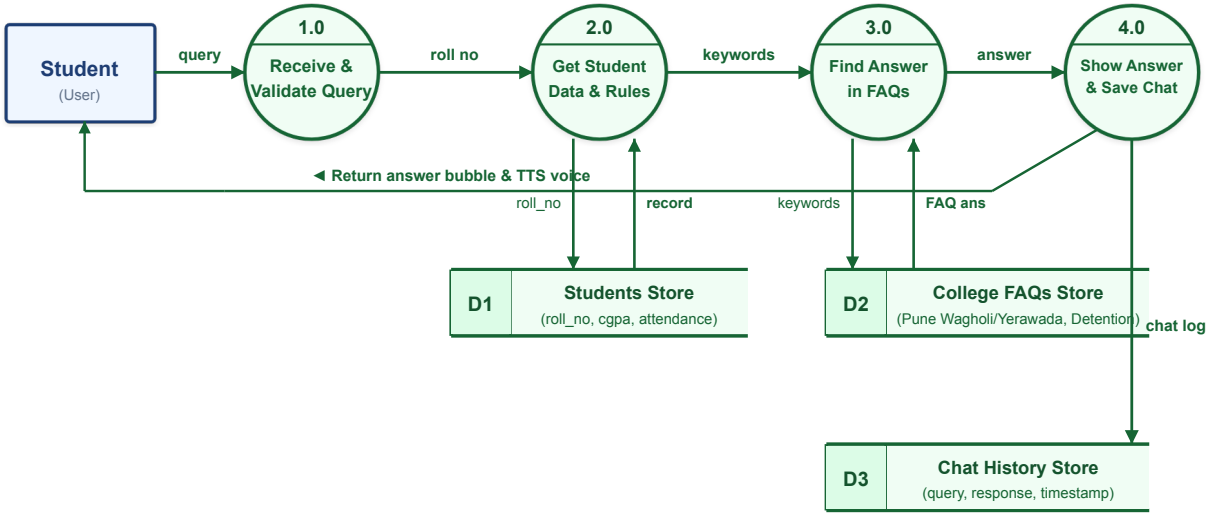
AI CHATBOT FOR COLLEGE

FACULTY REVISED SYSTEM ARCHITECTURE & DFD SPECIFICATION

✓ Context-Level DFD (Level 0) Included ✓ Standard Open-Ended Data Stores ✓ Circle Process Bubbles (DeMarco/Gane-Sarson)

Data Flow Diagram (Level 1: First-Level Process Decomposition & Open Data Stores)

First-Level Data Flow Diagram (DFD Level 1: Process Decomposition)



DFD Symbols and Their Uses (Standard Gane-Sarson & DeMarco Notations)

Symbol	Name	Standard Academic Definition	Implementation in Our Project
Student	External Entity	A person, department, or system outside the software boundary that feeds or receives data (Square/Rectangle)	**Student (User) & Administrator** Selects Pune campuses, requests attendance, updates FAQs.
1.0 Process	Process (Bubble)	An operation that transforms input data into output data. Standard notation is a circle with a process number and label.	**3.0 Find Answer in FAQs** Matches Wagholi & Yerawada campuses & 4-tier detention rules.
D1 Store	Data Store *(Open-Ended)*	A repository where data is held persistently at rest. **MANDATORY RULE: Must use open-ended parallel lines (NEVER a closed box).**	**D1 Students, D2 College FAQs, D3 Chat History** Stores (roll_no, cgpa, attendance, college_info, chat_history).
→	Data Flow	Directed solid arrow indicating the path of data in motion between entities, processes, and stores.	**query, roll_no, student record, FAQ answer** Directional arrows carrying specific payload packages.

Fig: First-Level Data Flow Diagram (DFD Level 1) — Process decomposition into 1.0 to 4.0 circle bubbles, standard open-ended parallel lines for data stores (D1 Students Store, D2 College FAQs Store, D3 Chat History Store), and full Gane-Sarson / DeMarco symbols table.